

A detailed study of the molecular beam flux distribution of MBE effusion sources

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We report experimental investigations of the thickness distribution of MBE-layers using effusion sources with crucibles of cylindrical and conical shape. The influence of various parameters—the filling level of the molten material, the evaporation rates, the inclination of the crucible, the crucible shape and, the distance between source and substrate—is studied in detail. Starting from the experimental data we demonstrate an estimation of the thickness distribution taking into account the interaction of the vapour particles.

1. Introduction

The angular distribution of the molecular flux from effusion cells used in molecular beam epitaxy (MBE) strongly influences the uniformity of epitaxial layers. The effusion sources are usually designed with tube-like crucibles. The large aperture of these crucibles allows us to produce a sufficient rate of flux already at lower temperature as compared with the Knudsen cell which is characterized by a small orifice and thermodynamical equilibrium inside the crucible. The angular distribution of a tube-like crucible crucially differs from the cosine distribution of the Knudsen cell and depends on the parameters of the tube (diameter, length, filling level, inclination angle, taper). The beaming effect of the tube and the deviation from the thermodynamical equilibrium have to be taken into account for calculation of the flux distribution. A detailed discussion of the effusion sources in the MBE technique is given by Herman and Sitter¹.

Earlier efforts^{2,3} to calculate the flux distribution of tube-like crucibles refer to Clausing⁴. These authors calculate the flow through tubes by an analytical approximation. Fujiwara⁵ has generalized Clausing's method involving inclined cylindrical crucibles. Curless⁶ and later Yamashita *et al*⁷ have given an algorithm to calculate the thickness distribution of epitaxial layers. At each point of the substrate they have summed the molecular flux coming from each point of the evaporating surface and from the reflecting walls of the crucibles assuming that the cosine distribution is valid at each point where a particle starts from. In a similar model Monte-Carlo simulations⁸⁻¹⁰ are successfully applied, taking into account the shape of the crucibles.

All authors mentioned have assumed the molecular flow regime, i.e. there is no interaction of the particles inside the crucibles (Knudsen number $\lambda/D \gg 1$, λ is the mean free path of particles, D is the typical size parameter of the crucible). As shown below, this assumption is not valid for the evaporation process in MBE with typical growth rates of about $1 \mu\text{m h}^{-1}$. Calculating the molecular beam flux for high rate nozzle sources

where the interaction of the particles has to be taken into account, Giordmaine and Wang¹¹ and Jackson *et al*¹² have applied a model using a fit parameter experimentally determined. Recently, Monte-Carlo simulations for $\lambda/D < 1$ were published^{13,14}.

Only some experimental data of the flux distribution from MBE effusion sources are available^{6,7,15}. The aim of this paper is to study the influence of (i) the filling level, (ii) the source temperature, i.e. the growth rate, (iii) the inclination of the crucible, (iv) the shape of the crucible, and (v) the distance between source and substrate on the uniformity of the epitaxial layers, in detail.

2. Experimental

The experiments were performed with bismuth having a low melting point (271°C) and a sufficient vapour pressure at not too high temperatures (10^{-2} – 10^0 Pa at 500–800°C)¹⁸. Graphite was used as the crucible material because it is easy to machine. Both crucibles, the cylindrical and the conical, had the same aperture diameter $D = 16.5$ mm. The thickness distribution was measured by monitoring the deposition rate in a plane perpendicular to the crucible axis. For this purpose a moveable quartz of a microbalance was scanned across the diameter of the plane (ϕ 96 mm) in different azimuths. The quartz position was registered by a potentiometer connected with a rotary feedthrough. The experimental arrangement is schematically drawn in Figure 1. The evaporation rate was controlled by the source temperature T measured with a Pt–Pt/Rh thermocouple and calibrated by observing the melting point of bismuth. The quartz was held at room temperature and we did not observe any drift of the deposition rate caused by thermal radiation from the sources. The distance R between the crucible aperture and the plane of registration was chosen as 115 mm. In a single experiment the distance was increased to 265 mm in order to determine the influence of the distance on the thickness distribution.

3. Results and discussion

Typical experimental data for two deposition rates are shown in Figure 2. The noise of the analogue output of the quartz

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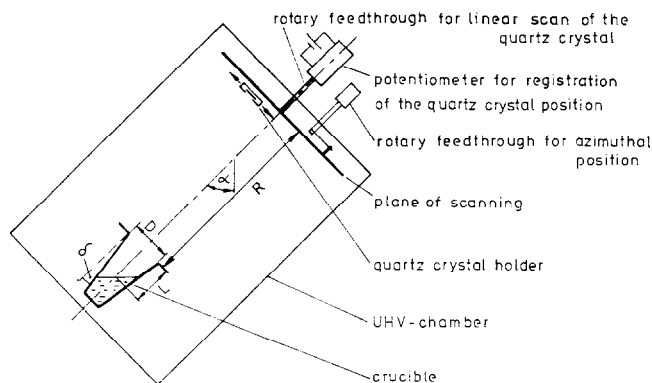


Figure 1. Scheme of the experimental arrangement for measurements of the thickness distribution ($\alpha = 0$ and 60° , $D = 16.5$ mm for both of the crucibles used, $\delta = 2.6^\circ$, $R = 115$ or 265 mm).

microbalance was about 0.1% for a deposition rate of $10 \text{ \AA s}^{-1}$ and about 4% at a rate of $0.1 \text{ \AA s}^{-1}$ illustrated by the bars in the lower curve. In the following the averaged experimental data are presented by solid lines in normalized representation.

Firstly, we have studied the dependence of the thickness distribution on the filling level of the crucible shown in Figure 3 for the conical crucible. The filling level is described by the parameter L/D where L means the distance of the surface of the molten material from the crucible aperture (see Figure 1). In the central part (diameter of about 30 mm) the thickness distribution has a plateau and does not depend on L/D . Outside this range the thickness distribution strongly depends on the filling level. In addition it was found that a measurable reduction in the absolute deposition rate with decreasing filling level can be observed only

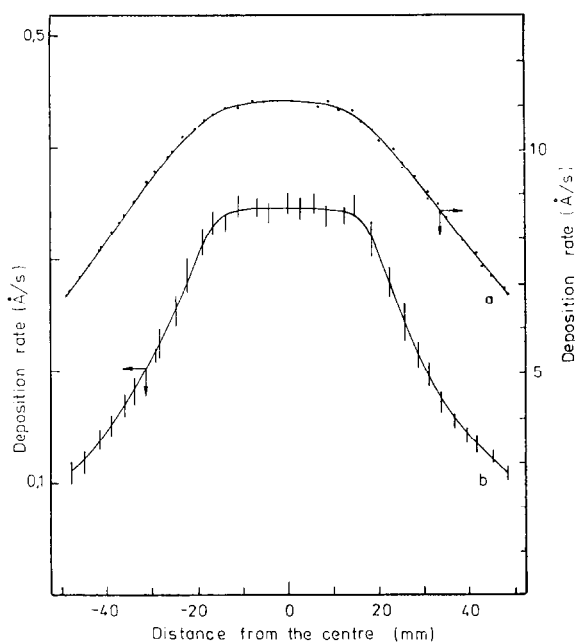


Figure 2. Deposition rate vs position at the substrate plane measured by a microbalance for the conical crucible ((a) $T = 723^\circ\text{C} \cong p_{\text{Bi}} = 4 \times 10^{-9}$ Pa, (b) $T = 573^\circ\text{C} \cong p_{\text{Bi}} = 8 \times 10^{-2}$ Pa, $L/D = 4$). The bars indicate the noise at low deposition rates. The solid lines are averaged fits.

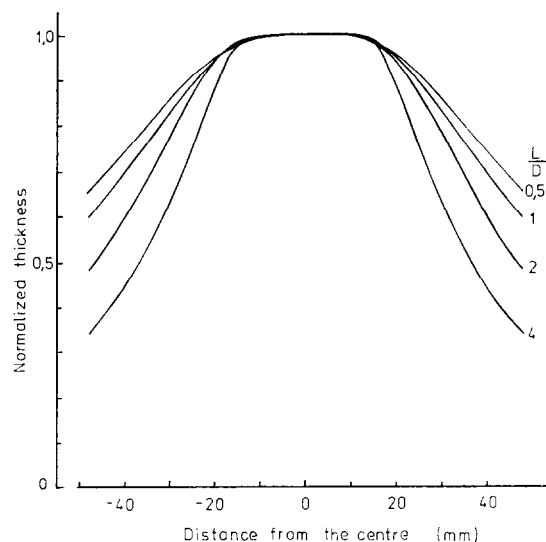


Figure 3. Normalized thickness distributions as measured for different filling levels L/D of the conical crucible ($T = 573^\circ\text{C} \cong p_{\text{Bi}} = 8 \times 10^{-2}$ Pa).

for $L/D > 3$. Exceeding this value a rapid reduction commences reaching 40% for $L/D = 4$.

Figure 4 shows the dependence of the thickness distribution on the source temperature and hence on the deposition rate. An increase of the source temperature results in a flattening of the distribution similar to an increase of the filling level.

In real growth processes, as different species are evaporated at very different rates (for example, dopants), the strong temperature dependence outside the central region shown in Figure 4 has to be considered. The ratio between the deposition rates of the different evaporants rises towards the wafer edge. We can estimate that lateral differences up to 20% in the doping level may follow for a 3 in. wafer.

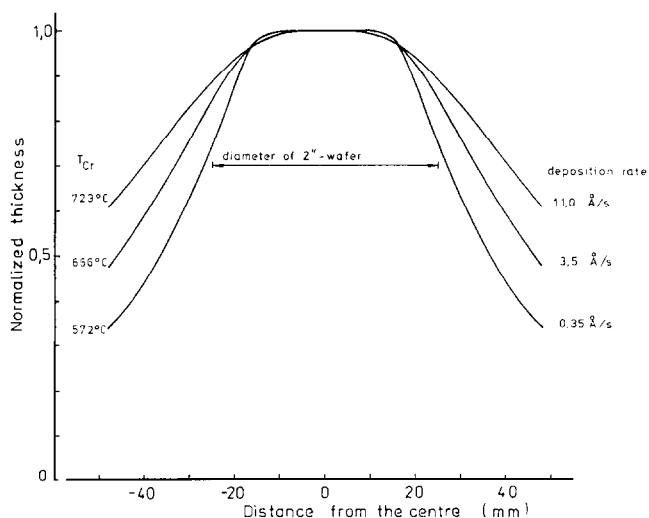


Figure 4. Normalized thickness distributions as measured for different temperatures T of the conical crucible at $L/D = 4$ ($T = 723^\circ\text{C} \cong p_{\text{Bi}} = 4 \times 10^{-9}$ Pa, $T = 666^\circ\text{C} \cong p_{\text{Bi}} = 8 \times 10^{-2}$ Pa, $T = 573^\circ\text{C} \cong p_{\text{Bi}} = 8 \times 10^{-2}$ Pa). The deposition rates are given for the centre of the distribution.

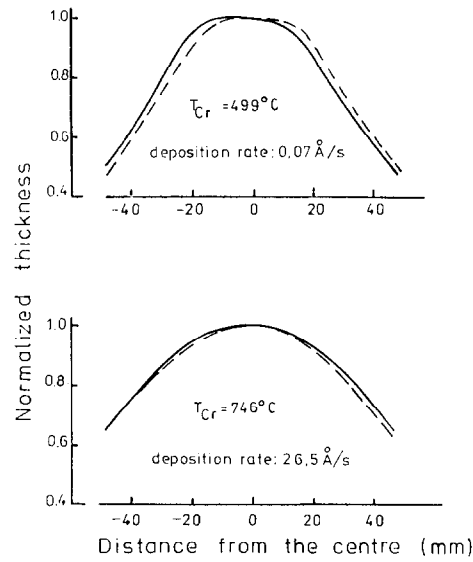


Figure 5. Thickness distributions for an inclined ($\alpha = 60^\circ$) conical crucible along the diameters for the azimuth $\tau = 0$ (solid line) and $\tau = 90^\circ$ (dashed line), $L/D = 4$.

In the case of molten material the flux distribution depends on the inclination of the source axis (angle α in Figure 1)¹⁰. We have studied the thickness distribution at $\alpha = 60^\circ$. In Figure 5 the thickness distributions measured in two azimuths ($\tau = 0-90^\circ$) are compared for different temperatures. There is only a small deviation from the rotation symmetry and at high evaporation rates ($\geq 10 \text{ Å s}^{-1}$) the distribution is rotation symmetrical.

For the homogeneity of the grown layers we have to take into consideration the shape of the crucible and the distance between source and substrate, too. Although the cylindrical and conical crucibles have the same characteristics according to the dependence on the growth rate and the filling level—there are significant differences in the thickness distribution (Figure 6). For

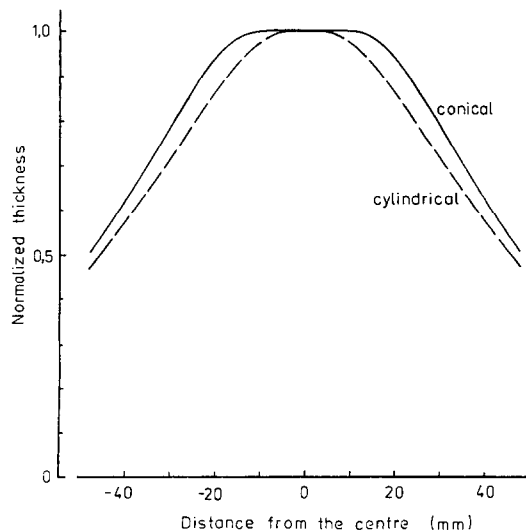


Figure 6. Comparison of the thickness distributions produced by the cylindrical and the conical crucibles ($T = 630^\circ\text{C}$, $L/D = 2$).

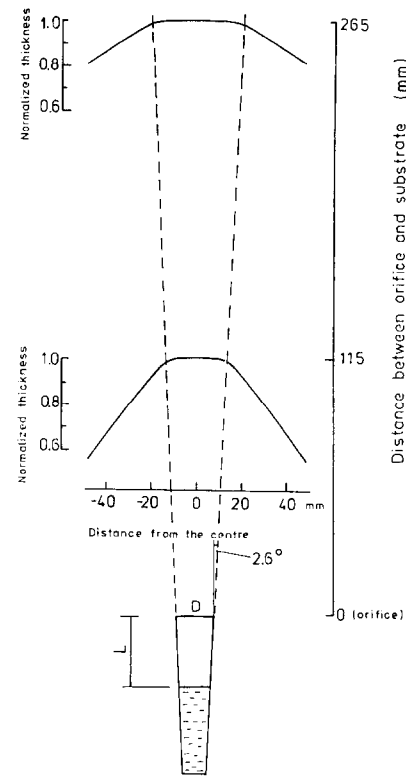


Figure 7. Thickness distributions for two distances between the source and the substrate (conical crucible, $L/D = 2$, $T = 593^\circ\text{C}$). The deposition rates in the centre were 10.0 and 1.9 Å s^{-1} for the distances 115 and 265 mm, respectively.

the conical crucible we get a wider plateau around the centre and therefore improved homogeneity. It is worth mentioning that at higher filling levels ($L/D < 0.5$) and/or higher growth rates ($\geq 10 \text{ Å s}^{-1}$) the influence of the shape of the crucible is vanishing.

The thickness distribution at two distances between the source and the substrate reveals a beaming effect (Figure 7). Quantitatively, the absolute deposition rate decreases as R^2 increases, where R is the distance between the crucible aperture and the substrate.

The dependence of the thickness distribution on the source temperature and the observation of the quasi-symmetry, despite an inclination of the source axis, imply that the interaction of particles inside the crucible cannot be neglected at practical growth conditions. For Bi we have found that the thickness distribution is independent of the temperature up to 550°C corresponding to a vapour pressure of $4 \times 10^{-2} \text{ Pa}$ ¹⁸ or to a Knudsen number $L/D > 12$. Under our experimental conditions ($R = 115$ mm), only for a deposition rate less than 0.2 Å s^{-1} can the intermolecular collision be neglected. In a MBE-system with a larger distance R this deposition rate is even smaller.

As mentioned above the Monte-Carlo simulation is appropriate to compute the thickness distribution for any MBE configuration but the incorporation of particle interactions still requires a parameter which can only be determined experimentally¹³.

If an angular distribution of flux $f(\Theta, \Phi)$ is available we can estimate the thickness distribution of any given MBE configuration or optimize the system design¹⁹. We calculate the

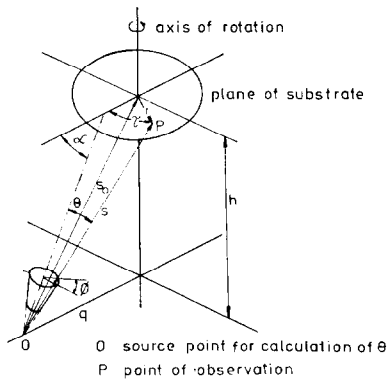


Figure 8. Evaporation geometry of a MBE-system (S_0 is the vector from the point source to the centre of the substrate and S to the point of observation).

angular distribution $f(\Theta, \Phi)$ using the measured thickness distribution $t_m(r)$. To define the angular distribution $f(\Theta, \Phi)$, it is necessary to start from a point source which gives the measured thickness distribution. The experimental data shown in Figure 7 imply that the point source has to be placed at the intersection of the projected crucible taper (see Figure 8). In a similar connection Saito *et al.*^{15,16} had demonstrated that this is valid at least for taper angles of the crucible between 1 and 5°. On the other hand, as the distance between the source and substrate is concerned one has to start from the aperture of the crucible as reported above (distance R). Using $f(\Theta, \Phi)$ the normalized thickness distribution $t(r)$ for a rotating substrate is given by^{8,17}:

$$t(r) = 1/\pi \int_0^\pi f(\Theta, \Phi) |S_0/S|^3 d\tau.$$

The parameters are explained in Figure 8. We can calculate with a rotation symmetric distribution $f(\Theta, \Phi) = f(\Theta)$ since the asymmetry of the thickness distribution in Figure 5 is small. Therefore we obtain $f(\Theta)$ from the measured distribution $t_m(r)$ as:

$$f(\Theta) = t_m(r) \cdot |\Delta S/\Delta S_0|^3.$$

As mentioned above the angle Θ has to be computed from the assumed point source. ΔS_0 and ΔS are the distances between aperture and substrate at S_0 and S , respectively (see Figure 8). For $t(r)$ we obtain:

$$t(r) = 1/\pi \int_0^\pi f(\Theta) h/S |\Delta S_0/\Delta S|^2 d\tau$$

(for h see Figure 8).

To demonstrate the influence of substrate rotation we have calculated the expected thickness distribution in a MBE-system using the conical crucible considered here and the geometry parameters of the VG V80H system (Figure 9). The homogeneity can be further improved by application of larger diameter D and larger taper angles δ .

4. Summary and conclusion

For growth rates normally used in MBE ($0.1\text{--}20.0 \text{ \AA s}^{-1}$) we have investigated the influence of (i) the filling level, (ii) the source temperature, i.e. growth rate, (iii) the inclination of the crucible, (iv) the shape of the crucible, and (v) the distance between

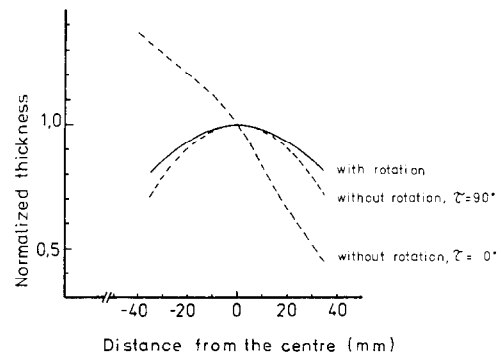


Figure 9. Thickness distributions estimated for the VG V80H MBE system (see Figure 8: $q = 160 \text{ mm}$, $h = 302 \text{ mm}$, $\alpha = 24^\circ$). The thickness distribution obtained with the conical crucible as shown in Figure 6 was used for the estimation ($T = 603^\circ \text{C}$, $L/D = 2$).

source and substrate on the thickness distribution of the deposited layers.

A high filling level as well as an increase of the growth rate result in a flattening of the thickness distribution. The influence of the inclination of the source is small at least up to inclination angles of 60° and can be neglected in practical MBE applications. A conical crucible gives a higher homogeneity of the layers than a cylindrical one if the central part of the flux distribution is used. It was verified that for tube-like evaporation sources the $1/R^2$ dependence of the deposition rate on the distance R between source aperture and substrate can be used with sufficient accuracy.

We can conclude from our experimental data that the interaction of the particles has to be taken into account for theoretical considerations for deposition rates higher than about $0.2 \text{ \AA s}^{-1}$, i.e. for Knudsen numbers $\lambda/D < 12$.

Using a single deposition rate scan across a plane perpendicular to the source axis it is possible to estimate the film thickness distribution for a given MBE configuration.

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